

Fig. 9: Components of the voltage divider appropriately wired

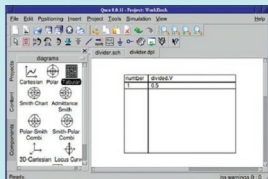


Fig. 12: Data display with tabular graph

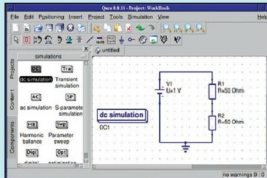


Fig. 10: Ground symbol as well as DC simulation in place

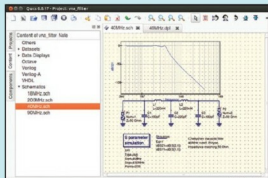


Fig. 13: Simple filter in qucs-qt4

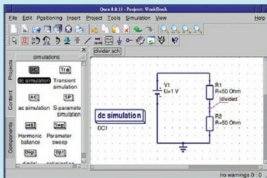


Fig. 11: Final voltage divider schematic

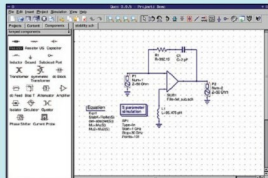


Fig. 14: Simple schematic area

sources), transmission lines (like microstrip, coaxial cable, twisted pair), nonlinear components (like ideal op-amp, transistors), digital components (like flip-flops), file components (like Touchstone files, Spice files), simulations (AC or DC analysis), diagrams (like Cartesian or polar plot) and paintings (such as texts, arrows, circles).

Each of the components can be placed on the schematic by clicking it once. Then move the cursor onto the schematic and click again to put it on its final position. You can right click to rotate the component into its final

position. You can even drag-and-drop the components.

DC simulation of a voltage divider

DC analysis is a steady-state analysis. It computes the node voltage as well as branch currents of the complete circuit. The circuit in Fig. 6 is for dividing the voltage of a DC voltage source according to the resistor ratio.

Wiring components. Connect the components appropriately using the wiring tool. Enable the wiring mode either by clicking the wire icon or by pressing the Ctrl + E shortcut.

Left clicking on the components' ports (small red circles) starts a wire and clicking on a second port finishes the wire. In order to change the orientation of the wire, right click it. You can leave the wiring mode by pressing Esc key.

For any analogue simulation (including the DC simulation), there is a reference potential required (for the nodal analysis). The ground symbol can be found in the Components tab in the lumped components category. The user can also choose the ground symbol icon or simply press the Ctrl + G shortcut. In the circuit in Fig. 7, the ground symbol is placed at the negative terminal of the DC voltage source.

Placing simulation blocks. The type of simulation

that is to be performed must also be placed on the schematic. You can do this by choosing the 'DC simulation' block, which is found in the Components tab in the simulations category.

Labelling wires. If you want voltage between the two resistors (the divided voltage) be output in the dataset after simulation, you need to label the wire. This is done by double clicking the wire and giving it an appropriate name. Wire labelling can also be done using the icon in the toolbar, by pressing the Ctrl + L shortcut or by choosing the Insert → Wire Label menu entry.